Chapter 1 Multiple-Choice Questions

Basic Concepts of Programming · 30 questions

Score: 0 / 30 · answered 0

1. Which statement best describes a computer program?

A. A physical part of the computer

B. A list of computer users

C. A set of explicit, unambiguous instructions that tells the computer what to do

D. A guess the computer makes

2. Writing, testing, debugging and maintaining source code is called...

A. Typing

B. Networking

C. Programming

D. Hardware design

3. Which is NOT a reason to study programming?

A. It removes the need to test your work

B. It develops logical thinking

C. It can automate repetitive tasks

D. It opens career opportunities

4. Which life cycle stage comes first?

A. Maintenance

B. Planning

C. Testing

D. Coding

5. A feasibility study checks whether...

A. a program has bugs

B. a flowchart is correct

C. a project is worth doing

D. a language is high-level

6. Which feasibility check asks whether users and management will accept the system?

A. Technical

B. Operational

C. Economic

D. Organizational

7. Requirements analysis focuses on...

A. what must be done, not how

B. installing hardware

C. writing the code

D. fixing bugs

8. Locating and fixing errors in a program is called...

A. Debugging

B. Documenting

C. Planning

D. Deploying

9. Testing the whole system on its real platform is...

A. System testing

B. Unit testing

C. Integration testing

D. Maintenance

10. Updating a program after the grading rules change belongs to...

A. Design

B. Maintenance

C. Requirements analysis

D. Planning

11. The three questions of problem analysis are about...

A. hardware, software and network

B. input, process and output

C. cost, time and size

D. author, date and title

12. A problem is best described as...

A. an error message

B. a programming language

C. the gap between the existing and the desired situation

D. a type of flowchart

13. Splitting a large problem into smaller parts is...

A. top-down design

B. bottom-up design

C. integration testing

D. debugging

14. Which is a characteristic of an algorithm?

A. Ambiguity

B. Randomness

C. Finiteness

D. Endlessness

15. An algorithm is independent of...

A. its input

B. any programming language

C. its output

D. its steps

16. A computer needs detailed step-by-step instructions because it is...

A. a heuristic-responding machine

B. an algorithm-responding machine

C. able to learn by itself

D. able to guess what we mean

17. Pseudocode is...

A. a type of memory

B. a compiled program

C. machine code

D. an informal, language-independent description of an algorithm

18. Which flowchart symbol marks Start and Stop?

A. Oval (terminal)

B. Diamond

C. Parallelogram

D. Rectangle

19. A parallelogram in a flowchart represents...

A. a decision

B. input or output

C. a calculation

D. a connector

20. A diamond in a flowchart represents...

A. a decision with two exits

B. a calculation

C. input or output

D. the end of the program

21. Why must every loop have a test?

A. To print results

B. To read data

C. To name variables

D. Otherwise it never ends

22. In the loop that adds 2 + 4 + ... + 30, the starting value of sum is...

A. 0

B. 2

C. 1

D. 30

23. What does this print? sum ← 0, i ← 1; while i <= 5: sum ← sum + i, i ← i + 1; print sum

A. 20

B. 5

C. 15

D. 10

24. Which language generation uses only 0s and 1s?

A. Third (high-level)

B. First (machine language)

C. Second (assembly)

D. Fourth

25. A program that translates assembly language into machine code is

a...

A. text editor

B. web browser

C. spreadsheet

D. assembler

26. Which of these is a high-level language?

A. Binary code

B. Machine language

C. Assembly language

D. C++

27. SQL is an example of a...

A. second-generation language

B. fourth-generation language

C. first-generation language

D. machine language

28. The paradigm “program = objects + messages” is...

A. functional

B. object-oriented

C. logical

D. imperative

29. Prolog follows which paradigm?

A. Unstructured

B. Imperative

C. Logic (declarative)

D. Object-oriented

30. Encapsulation means...

A. translating code to binary

B. bundling data with the functions that use it

C. drawing a flowchart

D. copying code between programs

Start again Show answer key